Psychology and Neuroscience 48th Annual Graham Goddard In-House Conference

June 07, 2024

Life Sciences Centre Room 4260-5260

IN-HOUSE SCHEDULE – FRIDAY, JUNE 7TH

TALKS

(All talks will be held in LSC Psychology, rooms 4260 & 5260 on the 4th-5th floor)

8:55 Opening Remarks LSC Psychology, Room 4260 on the 4th floor

---

9:00-10:30 Session 1 (Chair: Julie Blais) LSC Psychology, Room 4260 on the 4th floor - Panel on AI in Academia

---

Leslie Phillmore: View from the administration
Jennifer Stamp: Using AI as a course tool
Aaron Newman: Using AI to optimize research
Can Sozuer: Optimal LLM use and LLM use identification
Chris Holland: AI as a research tool and industry applications

10:30-11:00 Coffee Break - 4th Floor Lilyan E. White Study Centre

---

11:00-12:30 Session 2 (Chair: Gail Eskes) LSC Psychology, Room 4260 on the 4th floor

---

T1) 11:00	Deacon *How theory can inform practice and back again: a virtuous circle.*
T2) 11:15	Dolek *Are cueing effects really eliminated when accuracy is emphasized, and single masks eliminate uncertainty about target location?*
T3) 11:30	Klein *Assessing the Networks of Attention: ANT-Database and Portable Attention-Trip*
T4) 11:45	Eskes *Visual Scanning Training for Visuo-spatial Neglect*
T5) 12:00	McCormick *It's About Time: Distinguishing the Influence of Reflexive and Volitional Mechanisms of Temporal Attention*
T6) 12:15	Kaser *Bullying prevalence and associations with mental health problems among Canadian undergraduates during the COVID-19 pandemic*

12:30-2:00 Lunch & Poster Session - 4th Floor Lilyan E. White Study Centre

---

2:00-3:30 Session 3A (Chair: Shannon Johnson) LSC Psychology, Room 4260 on the 4th floor

---

T7) 2:00	Johnson *A year in review: The Centre for Psychological Health*
T8) 02:15	Shimizu *What should we be studying? Research priorities according to individuals with Sexual Interest/Arousal Disorder and their partners*
T9) 02:30	Irish *Juvenile Idiopathic Arthritis and Pain-Related Stigma*
T10) 02:45	Hoferek *Children's reading in digital and paper contexts: in which context is there higher levels of reading comprehension and why?*

T11) 03:00 Ethier-Gagnon *Trauma and cannabis cue-induced reward circuit alterations in cannabis users with trauma histories*

T12) 03:15 Stewart *Effects of trauma cue exposure and PTSD on affect and cannabis cognition in cannabis users with trauma histories: Use of expressive writing as an online cue-reactivity paradigm*

2:00-3:30 Session 3B (Chair: Ian Weaver) LSC Psychology, Room 5260 on the 5th floor

T13) 2:00 Semba *Sleep history-dependent astrocytic structural remodeling at synapses to wake-promoting orexin neurons in the lateral hypothalamus*

T14) 2:15 Mitchell *Prior short treatment of amblyopia in kittens reduces efficacy of later therapy.*

T15) 2:30 Li *Alzheimer's Disease associations with estrogen through changed methylation of ERα promoter regions and Aß42 binding*

T16) 2:45 Holland *Correlated but Not Connected: The Influence of Trust on Reliance Behaviour When Working with Automation*

T17) 3:00 Elassy *Distribution & Recruitment Patterns during Locomotion*

T18) 3:15 Drake *Cluster combo! Searching for synergy between mindfulness and cognitive training*

3:30-4:00 Coffee Break - 4th Floor Lilyan E. White Study Centre

4:00-5:15 Session 6 (Chair: Sean Mackinnon) LSC Psychology, Room 4260 on the 4th floor

T19) 4:00 Gazit Psychological interventions for teenagers with posttraumatic stress disorder: A scoping review

T20) 4:15 Hill It's the Little Things in Life: Enjoyment of Different Types of Personal Projects

T21) 4:30 Mackinnon Poor grades in statistics classes longitudinally predict increased anxiety and worsened attitudes towards statistics education

T22) 4:45 Hill Focused, flourishing, but not in flow: Achievement strivers' experiences of competence, flow, and well-being during personally expressive activities

T23) 5:00 Monchesky Age, but not food-deprivation, affects hippocampal neuronal recruitment in the domesticated pigeon (Columba livia domestica)

Concluding Remarks

POSTERS

(All posters will be presented from 12:30 pm to 2 pm in the Lilyan E. White Study Centre on the 4th floor)

1	Kyle Roddick
2	Bianca Matthews
3	Mya Dockrill
4	Betsy McCord
5	Grace Schwenck
6	Ry Lewis
7	Lan Dang
8	Can Sozuer

9	Isabella Boossom
10	Lindsay Aubin
11	Ali Dyack
12	Vanessa Ritsema
13	Mark ten Haaf
14	Laura Elliott
15	Moomina Raja
16	Emily McDonald (Student)
17	Lamisa Rahman (Student)
18	Sam Hawkins
19	Basmah Hendy
20	Letícia Aihara de Oliveira
21	KC Collings
22	Pristine Garay
23	Kameryn Whyte
24	Prem Kundhi
25	Celia Glenham
26	Emma Reeves

TALK ABSTRACTS

Monchesky, Ainsley

TALK TITLE: Age, but not food-deprivation, affects hippocampal neuronal recruitment in the domesticated pigeon (Columba livia domestica)

AUTHORS: Ainsley N. Monchesky, Broderick M. B. Parks, Casandra Hayar, Jason R. Treberg, Debbie M. Kelly, Leslie S. Phillmore

ABSTRACT: The hippocampus, a brain region important for learning and memory, plays an inhibitory role in the stress response. Despite this structure being relatively conserved across evolution, mammals and birds navigate different environments, suggesting that mechanisms of neural plasticity, including neurogenesis, have allowed for environmental adaptations in hippocampus-based spatial cognition. Additionally, age impacts the mammalian and avian brain differently: mammalian neurogenesis declines with age whereas avian neurogenesis is brain-wide and life- long. Metabolic stress, like food deprivation, also decreases hippocampal neurogenesis, further suggesting that it plays a role in learning, memory, and the stress response. Therefore, my project examined the relationship between food deprivation, age, and neurogenesis in the hippocampus of the domesticated pigeon (Columba livia domestica), an avian spatial specialist. Subjects (n = 22) underwent a food-deprivation manipulation prior to sacrifice. Brain tissue was stained for doublecortin, an endogenous marker of neurogenesis; I quantified immunolabelling in three subregions of the hippocampus (DL, DM, and TrV). I found that TrV had the most neurogenesis across both age groups, however recruitment to DM and DL differed with age. The food deprivation manipulation did not affect neurogenesis. Overall, our findings show that pigeons may be resilient to metabolic stress, and neuronal recruitment may differ with age.

Kaser, Alanna

TALK TITLE: Bullying prevalence and associations with mental health problems among Canadian undergraduates during the COVID-19 pandemic

AUTHORS: Kaser, A. C. (presenter), Lambe, L. J., Yunus, F. M., Conrod, P. J., Hadwin, A. F., Keough, M. T., Krank, M. D., Thompson, K., Stewart, S. H.

ABSTRACT: The COVID-19 pandemic impacted educational contexts and mental health and has been associated with heightened adolescent cyberbullying; however, bullying research at university is limited. We examined gender and pandemic-related differences in cyber- (online) and traditional (in-person) bullying among 1435 first-year undergraduates and bullying associations with internalizing (anxiety; depression) and externalizing problems (alcohol use; risky behaviour). For overall bullying prevalence, victimization (47.1%) exceeded perpetration (26.2%) and traditional bullying (51.7%) exceeded cyberbullying (18.2%). Women had higher odds of victimization compared to men. Surprisingly, Fall 2022 (low COVID-19 restrictions) students had higher odds of cyber-victimization and -perpetration than Fall 2021 (high COVID-19 restrictions) students. Positive associations were identified for internalizing outcomes with traditional and cybervictimization, alcohol use with traditional perpetration, and risky behaviour with all forms of bullying. Bullying persists post-pandemic and has concerning associations with emerging adults' mental health, warranting continued monitoring and support during the transition to university.

Irish, Blair

TALK TITLE: Juvenile Idiopathic Arthritis and Pain-Related Stigma

AUTHORS: Blair N. Irish 1-2 ;Dr. Christine T. Chambers1-3 ;Dr. Justine Dol 2 ;Dr. Adam H. Huber 3-5 ;Yvonne N. Brandelli 1-2

1 Department of Psychology and Neuroscience, Dalhousie University, Halifax, NS
2 Centre for Pediatric Pain Research, IWK Health, Halifax, NS
3 Department of Pediatrics, Dalhousie University, Halifax, NS
4 Division of Rheumatology, Dalhousie University, Halifax, NS
5 Department of Rheumatology, IWK Health, Halifax, NS

ABSTRACT: Juvenile Idiopathic Arthritis (JIA) is a chronic inflammatory condition associated with pain in childhood. Youth with JIA have reported experiencing negative judgement from others in response to their feelings of pain (i.e., pain-related stigma). This can negatively affect their physical and psychological health. How pain-related stigma impacts individuals' health outcomes is partly determined by their reactions to these experiences. This study will explore how youth with JIA and their caregivers react to pain-related stigma. To accomplish this, 15 youth (ages 13-18) with JIA and 15 caregivers of a youth with JIA will complete separate semi-structured interviews with questions asking about their (or their child's) experiences of pain-related stigma and how they reacted to these experiences. The interview data will be analyzed using reflexive thematic analysis. This will result in separate themes describing how youth with JIA and caregivers of youth with JIA react to pain-related stigma.

Holland, Chris

TALK TITLE: Correlated but Not Connected: The Influence of Trust on Reliance Behaviour When Working with Automation

AUTHORS: Christopher Holland, Grace Perry, Heather Neyedli

ABSTRACT: Trust plays a crucial role in human-automation collaboration, with imbalances leading to performance issues (Lee & Moray, 1992; Stowers et al., 2000). Trust's impact on user behavior is influenced by factors such as self-confidence, automation capabilities, and workload. Our lab's previous work used a binary choice task to show that trust in automation is dynamic and influenced by initial performance. Automation with moderate reliability can build trust over time, whereas initially unreliable systems struggle, even if reliability improves (Rittenberg et al., 2024). Despite reported distrust, user performance often aligns with automation reliability, suggesting distinct yet intertwined constructs of trust and reliance. This study aims to understand trust and reliance dynamics with varying automation reliability, directly assessing reliance behavior and capturing self-reported self-confidence. We also explore the effects of feedback, task order, and initial reliability on trust calibration, aiming to enhance the design of future automated systems and interfaces.

McCormick, Colin

TALK TITLE: It's About Time: Distinguishing the Influence of Reflexive and Volitional Mechanisms of Temporal Attention

AUTHORS: Colin R. McCormick & Raymond M. Klein

ABSTRACT: In a noisy and dynamic world, the ability to modulate arousal and predict the temporal structure of events is critical to improve how efficiently we process and respond to information. Despite over a century of research on temporal attention, there remains inconsistency in the naming of mechanisms, which involves the confounding of independent modes of allocation. Relatedly, there is also uncertainty regarding the stage of processing which is affected. In this talk, I will present the results from two distinct temporal attention paradigms, which highlight distinctions between voluntary and reflexive modes of temporal attention. I will then explore how the converging evidence from these studies align with prior theories of temporal attention.

Johnson, Debbie

TALK TITLE: A year in review: The Centre for Psychological Health

AUTHORS: Shannon Johnson, Alissa Pencer, Debbie Emberly

ABSTRACT: The Dalhousie Centre for Psychological Health (CPH) is a collaboration between the Office of Addictions and Mental Health (OAMH) and Dalhousie University's Clinical Psychology PhD Program. CPH is dedicated to creating a supportive learning environment in which clinical psychology students can gain experience in conducting assessments and delivering treatment with individuals from diverse and marginalized backgrounds. Our aim is to increase access to psychological services, by providing low barrier and evidence-based mental health care, while also meeting student training needs. The CPH offers assessment and intervention services, including individual, couples, and group therapy.

Mitchell, Donald

TALK TITLE: Prior short treatment of amblyopia in kittens reduces efficacy of later therapy.

AUTHORS: Donald Mitchell and Kevin Duffy

ABSTRACT: Some clinical reports suggest that interventions to treat amblyopia in children may be less effective if they had received prior unsuccessful treatment. This possibility was explored in an animal model of amblyopia by use of conventional patching therapy, whereby the "good" or non-amblyopic eye is occluded to force use of the amblyopic eye. Deprivation amblyopia was induced in one eye of 10 kittens that had one eye occluded for a week at 3 weeks of age. The efficacy of treatment by occlusion (patching) of the non-amblyopic eye for 2-3 weeks at 5-6 weeks of age was compared between two groups of animals. Whereas one group of animals (the control group) received this treatment alone, for animals in the second group the same treatment was preceded by a very short (3 day) period of such occlusion of the non-deprived eye at around 4 weeks. The mean visual acuity recovered by the deprived eye of animals from the control group was 2 ½ times better than that observed in the animals that received two periods of patching treatment.

Eskes, Gail

TALK TITLE: Visual Scanning Training for Visuo-spatial Neglect

AUTHORS: Sierra Gaudreau, Chris Holland, Lucas Wan, Dr. Ya-Jun Pan, Dr. Heather Neyedli

ABSTRACT: Introduction: Visuo-spatial neglect (neglect) is a common attentional disorder after right hemisphere stroke characterized by difficulties with orienting, detecting, or responding to contralesional information. Visual scanning treatment (VST) is recommended, but effective treatment requires intensive practice for at least 10-15 hours, and thus it is difficult to implement with short stays and limited hospital resources.

Objectives: We aimed to examine the feasibility and effectiveness of an online VST program (SpyTrain) for neglect post stroke.

Methods: VST was delivered to 4 patients with neglect and 1 control using a non-concurrent multiple baseline single-case experimental design. Outcomes were analysed with Tau-U, a nonparametric test of non-overlap and trend.

Results: Findings were positive for feasibility and effectiveness on primary and transfer measures and will be discussed.

Conclusions: If this online method of VST continues to be feasible and effective in our ongoing research, it would provide an accessible form of treatment for a common and chronic disorder, improving independence and quality of life after stroke.

Deacon, Helen

TALK TITLE: How theory can inform practice and back again: a virtuous circle.

AUTHORS: Helene Deacon, Sofia Giatzitzidou

ABSTRACT: Children's books contain many complex words, and these can be a barrier to their ability to understand what they read. One of the primary ways in which words are complex is that they are made up of roots and affixes, also known morphemes. I review our recent theory on how children use morphemes to help them in their reading. I also show recent data from several large-scale longitudinal study testing these ideas, and how we can apply these data to instruction.

Shimizu, Justin

TALK TITLE: What should we be studying? Research priorities according to individuals with Sexual Interest/Arousal Disorder and their partners

AUTHORS: Shimizu JPK, Bergeron S, Schwenck GC, Huberman JS, & Rosen NO.

ABSTRACT: Objective: Inclusion of patient perspectives may help to reduce misalignments between the research priorities of patients and researchers, however, the priorities of women and gender diverse individuals with Sexual Interest/Arousal Disorder (SIAD) and their partners remains unclear. The purpose of this study was to identify the research priorities of women and gender diverse individuals with SIAD and their partners.
Methods: Couples coping with SIAD responded to an open-ended question asking them to list the top 3 things they think are important for SIAD researchers to focus on. A team-based qualitative thematic content analysis was conducted to identify themes.
Results: Analysis of 1429 responses revealed 6 main themes: general causes, general treatment, and coping, biophysiological, relationship, psychological, and environmental contextual. Additionally, 4 sub-themes were identified within the latter 4 of the main themes: general, causes, treatment, and impact.
Conclusion: Findings suggest that individuals with SIAD and their partners have similar and heterogeneous research priorities that are consistent with a biopsychosocial approach.

Elassy, Kareem

TALK TITLE: Distribution & Recruitment Patterns during Locomotion

AUTHORS: Kareem Elassy (PI: Ying Zhang)

ABSTRACT: During development, spinal interneurons (IN) split into distinct populations; one of which being the diverse V3 IN population which are critical for interlimb coordination during locomotion. Efforts to further subdivide these according to their task-dependent recruitment and topographic locations have been limited to the higher lumbar region in the spinal cord. We took an exploratory approach to identify distributions of V3 IN populations in all spinal segments in mice and compare them to previously classified clusters in the lumbar region. Using c-Fos as a marker for neuronal activity, we also investigated V3 IN speed-dependent recruitment patterns. V3 IN clusters were found to have distinct distributions between spinal segments. The proportion of active V3 INs did not differ between high and low locomotor speeds, however, at higher speeds, V3 IN activity was more localized suggesting a more specific reorganization of the circuitry for higher locomotor demands.

Semba, Kazue

TALK TITLE: Kazue Semba[1,2,] Tatjana Golovin[1], Chantalle Briggs[1], Joan Burns[1], Sayuri Hatada[3], Samuel Deurveilher[1], Yoshiyuki Kubota[3,4]

1 Departments of Medical Neuroscience, Dalhousie University
2 Department of Psychology & Neuroscience, Dalhousie University
3 Natl. Inst. Physiol. Sci. (NIPS), Okazaki, Japan
4 RIKEN Center for Brain Science, Wako, Japan
AUTHORS: Katie Hoferek, S. Hélène Deacon

ABSTRACT: Sleep-wake cycles are regulated by alternate activation of sleep- and wake-promoting neurons. Increasing evidence, however, indicates that astrocytes, formerly considered 'housekeeping cells,' are in fact important partners of neurons in various brain functions and behaviours. We previously showed that astrocytes dynamically regulate excitatory transmission to wake-promoting orexin (ORX) neurons according to sleep history, by altering the distribution of glutamate transporter-1 thereby controlling presynaptic inhibition. To test the hypothesis that this synaptic plasticity results from structural remodeling of perisynaptic astrocytic processes, we used serial electron microscopy combined with immunoconfocal microscopy. We found that astrocytic processes at synapses to ORX neurons physically withdraw from synapses following 6 h of SD. We propose that perisynaptic astrocytic restructuring is a cellular mechanism for sleep homeostasis.

Ethier-Gagnon, Mikaela

TALK TITLE: Trauma and cannabis cue-induced reward circuit alterations in cannabis users with trauma histories

AUTHORS: Ethier-Gagnon, M., Barret, S.P., Romero-Sanchiz, P., Helmick, C., McAllindon, D., Tibbo, P., Crocker, C., Good, K., Arenella, P., Rudnick, A., Cosman, T., Trottier, S.-J., DeGrace, S., Collins, P., Girgulis, J., Snooks, T., & Stewart, S. H.

ABSTRACT: Trauma histories increase risk for excessive and problematic cannabis use. The trauma-cannabis link may be related to conditioned craving to trauma cues which arises through classical and operant conditioning; underlying brain mechanisms remain unknown. N=23 cannabis users with trauma histories (Mage=33.5, 69.6% female) were presented with personalized trauma, cannabis, and neutral cues in random order using a cue reactivity paradigm in the MRI. Participants were assessed for subjective craving and emotional responses. Functional connectivity (FC) between striatal, cortical, and limbic regions was measured during cue presentation. Trauma and cannabis cues affected craving and emotions in expected ways. Trauma cues increased FC within the striatum and between the striatum and cortical regions, but decreased FC between cortical regions and dorsal striatum-limbic connectivity compared to other cues. Cannabis cues increased dorsal striatum-cortical connectivity but decreased overall cortical connectivity compared to neutral cues. Findings highlight the potential neural mechanisms underlying the trauma-cannabis link.

Klein, Raymond

TALK TITLE: Assessing the Networks of Attention: ANT-Database and Portable Attention-Trip

AUTHORS: Raymond M Klein, Colin McCormick, Swasti Arora

ABSTRACT: As put forward in Posner's taxonomy, attention refers to three basic neuro-cognitive processes: alerting, orienting, and executive control. Developed by Posner & colleagues over two decades ago the Attention Network Test (ANT) provides a convenient measure of the efficacy the networks that mediate these functions. Since then, the ANT and its variants have been so widely used to measure attention networks that we developed a publicly accessible database of all studies using any version of the ANT to answer any question. We have also developed a more engaging, game-like version of the ANT, AttentionTrip, which has recently been ported to the iPad. In this presentation, we will illustrate the value of the ANT-Database by describing the studies which have cited using it, and we will demonstrate the ways AttentionTrip can be used (and has been used).

Drake, Richard

TALK TITLE: Richard S. Drake, Amy E. Maxwell, Tammy L. Lamb, David Whitehorn, K. Ryan Wilson, Gail A. Eskes

AUTHORS: Raymond M Klein, Dawson Sutherland, Austin Hurst

ABSTRACT: Computerized cognitive training (attention process training) and mindfulness meditation practice (attention state training) are promising avenues for cognitive enhancement given their accessibility (e.g., at-home, cost-effective, available for all ages). Tang and Posner (2009) positioned the two approaches as complementary but orthogonal methods that help an individual avoid "two extremes of the untrained mind": mental fatigue (decreased through attention process training) and mind wandering (decreased by attention state training). Although Tang and Posner imply that combining both training types might confer additional benefits (by reaching an "attention balance state"), dual intervention studies are uncommon. In this talk, we report our in-progress work examining the specific and combined effects of guided, mindfulness meditation and adaptive n-back training. To improve our power and assess the stability of the effects, we analyzed three datasets that used similar methodologies across in-person and at-home (online) formats. We will discuss trends, implications and applications, and next steps.

Sean, Mackinnon

TALK TITLE: Exploring the Role of Attentional Withdrawal in Item-Method Directed Forgetting Through Visual Search

AUTHORS: Ryan Lewis, Tracy L. Taylor

ABSTRACT: We embedded a serial and pop-out visual search into the study phase of an item method-directed forgetting paradigm to assess attentional availability following instructions to remember (R) or forget (F). Study words were presented to the left or right of a central fixation (Experiments 1-2) or at center (Experiments 3-5), followed by a visual (Experiments 1-4) or auditory (Experiment 5) memory instruction, and then by the search display. Although the evidence did not support a differential withdrawal of attention following F compared to R instructions, Experiment 5 replicated slower response times to visual targets that followed F compared to R instructions (e.g. Fawcett and Taylor, 2008; 2012) - consistent with the view that intentional forgetting engages an active cognitive process. We speculate that this active process might be a non-attentional inhibitory mechanism that is employed to further reduce the encoding of F items alongside selective rehearsal of R over F item.

Sean, Mackinnon

TALK TITLE: Poor grades in statistics classes longitudinally predict increased anxiety and worsened attitudes towards statistics education

AUTHORS: Sean P. Mackinnon, Sean M. Alexander, Robert A. Cribbie, Gordon L. Flett, & Taylor G. Hill

ABSTRACT: Students report anxiety and lack of confidence in statistics classrooms. Because statistics classes can be difficult, early negative feedback (e.g., poor grades) might intensify anxiety and increase negative attitudes towards statistics. In the current study, we test whether statistics class grades predict future attitudes and anxiety towards statistics education. Participants included students taking a statistics class at time 1 ($N = 423$). We used a 2-wave longitudinal survey design, where participants completed surveys during the first month of classes and ~4 months later once grades were released. We measured statistics anxiety, statistics attitudes, and self-reported statistics grades. Statistics class grades were negatively related to statistics anxiety and attitudes towards statistics at time 2, even when controlling for baseline levels of outcomes at time 1. That is, students who received poor grades experienced increased anxiety and worsened attitudes towards statistics once the class was completed. Findings suggest early setbacks in statistics education (i.e., poor grades) may intensify negative attitudes towards statistics education which may lead students to avoid further study of statistics. Instructors might consider increased use of formative assessments and active attempts to reduce anxiety in the classroom to keep students engaged in statistics education.

Stewart, Sherry

TALK TITLE: Effects of trauma cue exposure and PTSD on affect and cannabis cognition in cannabis users with trauma histories: Use of expressive writing as an online cue-reactivity paradigm

AUTHORS: Effects of trauma cue exposure and PTSD on affect and cannabis cognition in cannabis users with trauma histories: Use of expressive writing as an online cue-reactivity paradigm

ABSTRACT: Conditioned memory associations between trauma cues, distress, and cannabis use may contribute to PTSD-cannabis use disorder (CUD) comorbidity. These associations can be studied experimentally by manipulating trauma cue exposure in a cue-reactivity paradigm (CRP). We examined CRP condition (trauma, neutral) and PTSD group (likely PTSD+, PTSD-) effects on affective and cognitive outcomes using an online expressive writing CRP. N=202 cannabis users with trauma histories (43.6% male; M age=42.9) completed a validated PTSD measure and were randomized to trauma or neutral expressive writing. Then they completed validated measures of affect, cannabis craving, and memory accessibility of cannabis information. Hypothesized main effects of expressive writing condition were found for negative affect, expectancy cannabis craving, and memory accessibility of cannabis information, and of PTSD group for negative affect and all cannabis craving dimensions. Interventions focused on reducing negative affect and expectancy craving to trauma cues may prevent/treat CUD among cannabis users with PTSD.

Li, Shuya

TALK TITLE: Alzheimer's Disease associations with estrogen through changed methylation of ERα promoter regions and Aß42 binding

AUTHORS: Shuya Li, Dr Ian Weaver

ABSTRACT: Alzheimer's disease (AD), a neurodegenerative disorder associated with Aß42, has gender-linked risks, implicating estrogen (E2), a neuroprotective hormone. The Weaver lab previously showed 5x-FAD mice, an AD model, had decreased methylation of ERα promoter region and increased Aß42 binding. This study investigated these results' applicability to humans and hypothesized that lower ERα expression in AD patients is associated with increased Aß42 binding, and decreased DNA methylation. Using twelve human frontal-cortex brain samples, ERα promoter methylation was determined using bisulfite pyrosequencing, and Aß42 binding, using ChIP-PCR. Of the 23 ERα promoter region methylation sites analyzed using bisulfite pyrosequencing, non-AD patients had increased methylation percent compared to AD patients (p<0.001). ChIP PCR of 5mC, Aß42 DNA interaction patterns found increased Aß-42 binding in the AßID region of AD patients (p<0.001). Methylation and Aß binding patterns suggest Aß42 involvement in ERα regulation, encouraging additional research of ERα mRNA transcript and E2 protein abundance.

Dolek, Simal

TALK TITLE: Are cueing effects really eliminated when accuracy is emphasized, and single masks eliminate uncertainty about target location?

AUTHORS: Şimal Dölek[1], Seema Prasad[2], John Christie[1], Raymond Klein[1].

1 Department of Psychology & Neuroscience, Dalhousie University, Halifax, NS, Canada
2 Carl Gustav Carus University Hospital, Technical University of Dresden, Dresden, Germany

ABSTRACT: The present study examined the influence of spatial pre-cuing on the accuracy and speed of masked target identification, with the single target covered by either a single mask on the target location or multiple masks

on all possible target locations. We analyzed how invalid versus valid cues affected participant reaction time (RT) and accuracy. Additionally, we introduced a speed-stress manipulation directing one participant group to favor response quickness, while the other was directed to prioritize accuracy without time constraints. In Experiment 1, clear cueing effects were found in RT and accuracy across all conditions. As Experiment 1 presented homogenous blocks of either single or multiple masked trials, a subsequent experiment randomized single and multiple mask trials within each block. This approach aimed to extract the true effects of cueing, detached from participants' strategic adjustments based on the higher informativity of the single mask compared to the pre-cue. Confirming initial findings, the follow-up experiment revealed consistent cueing effects across almost all conditions, in terms of both RT and errors. While our results support the signal enhancement theory of cueing effects, we also consider the potential of a hybrid model involving noise reduction and signal enhancement as the basis of this phenomena.

Gazit, Tamar

TALK TITLE: Psychological interventions for teenagers with posttraumatic stress disorder: A scoping review

AUTHORS: Tamar Gazit[1,2], Sydney Walker[2,] Patrick J. McGrath[1,2].

1 Department of Psychology and Neuroscience, Dalhousie University, Halifax, Nova Scotia, Canada
2 Centre for Research in Family Health, IWK Health Centre, Halifax, Nova Scotia, Canada

ABSTRACT: A scoping review was conducted to examine psychological interventions for treating posttraumatic stress disorder (PTSD) in teenagers. The review aimed to identify relevant studies, research methodologies, and knowledge gaps. We retrieved candidate studies from three databases on July 5, 2023, and screened citations using Covidence. English-language studies of psychological interventions for teenagers (ages 13-19) with PTSD were included. Out of 20,288 retrieved citations, 48 studies met the inclusion criteria. The most frequently reported interventions included Trauma-Focused Cognitive Behavioural Therapy (n=10), Prolonged Exposure (n=8), Cognitive Processing Therapy (n=5), and Narrative Exposure Therapy (n=4). Seventeen studies reported on unique interventions. Study designs varied, including randomized controlled trials (n=27), and quasi-experimental studies (n=10). Reported findings indicate that these interventions generally reduced PTSD symptoms. Gaps were noted in digital delivery methods, mechanisms of change, and interventions for less common traumas. These findings highlight critical areas for future research, particularly the need for additional empirical evaluation of both established and emerging treatment options for this vulnerable population.

Hill, Taylor

TALK TITLE: It's the Little Things in Life: Enjoyment of Different Types of Personal Projects

AUTHORS: Taylor G. Hill, Emma C. Coughlan, Sean P. Mackinnon

ABSTRACT: Many positive psychology interventions aim to improve happiness through engagement in simple everyday activities via intrinsic rewards. Personal projects are goal-directed activities that offer a way to study the intentional pursuit of happiness. This exploratory study identifies the types of projects that people engage in, and the dimensions predict hedonic well-being (enjoyment). Participants (N = 327) completed a cross-sectional survey. Two coders thematically coded projects; we used linear mixed models to identify which project types and dimensions uniquely predict enjoyment. People engaged in various types of activities which were enjoyed to different extents - they tend to experience greater enjoyment when projects encourage autonomy, control, likelihood of success, progress, absorption, low difficulty, and low challenge. Knowledge on which activity characteristics are linked to well-being can inform tailored positive psychology programming. Overall, people tend to find activities which are relatively easy and where they make a lot of progress more enjoyable, indicating simple daily activities are one way to intentionally prioritize daily well-being.

Hill, Taylor

TALK TITLE: Focused, flourishing, but not in flow: Achievement strivers' experiences of competence, flow, and well-being during personally expressive activities

AUTHORS: Taylor G. Hill, Johanna V. Loock, Sean P. Mackinnon

ABSTRACT: One route to increasing well-being is pursuing activities which suit one's personality strengths. Achievement strivers tend to organize their behaviors in ways that promote goal attainment and well-being. We tested the hypothesized process that achievement striving would lead to increased well-being over time through feelings of competence and flow. Students (N = 346 at Time 1; N = 244 at Time 2) completed an online survey measuring personality, personally expressive activities, basic psychological need satisfaction, flow, and well-being at two timepoints, about 4 months apart. We used cross-sectional and longitudinal serial mediation models to test our hypothesis with eudaimonic and hedonic well-being. Achievement striving was positively correlated to competence and well-being, but the indirect effects did not show that well-being is boosted by feeling competent and in flow during personally expressive activities, cross-sectionally or longitudinally. While achievement strivers tended to feel happy and competent at personally expressive activities, the mechanistic pathway to well-being is not yet clear. Future studies might recruit larger sample sizes and utilize smaller time lags.

POSTER ABSTRACTS

All Posters Will Be Presented From 12:30 pm to 2 pm in the Lilyan E. White Study Centre on the 4th floor.

Dyack, Ali

POSTER TITLE: Are You Wearing Your Clothes or Are Your Clothes Wearing You? The Roots of Children's Clothing-Based Inferences About Group-Level Knowledge

AUTHORS: Alexandra Dyack, Rebeka Workye, Marise Barsoum, Emma Reeves, Victoria Aragon, Drew Weatherhead

ABSTRACT: Clothing is a powerful social marker that allows adults and children to implicitly gather information about others such as group membership, occupation, and social status. Recent research shows that children make inferences about people's knowledge based on their clothing, meaning they can deduce a group of people wearing the same clothing are likely to share similar knowledge. However, this research does not speak to where children are attributing group-level knowledge. It is possible that they believe knowledge comes within an individual or that it is attached to clothing, such that a person gains knowledge when they put on certain clothes and loses it when the clothes are removed. The current study assessed 5-8-year-old children's beliefs about knowledge when clothing was manipulated, by asking whether an individual retained previous knowledge and gained new knowledge upon changing her clothes. On average, participants believed an individual retained her previous knowledge after changing her clothing, but were unsure if she gained knowledge associated with her new clothes. These results support person-specific knowledge attribution, where children believe knowledge is an internal property that persists across external changes. Early on, children make complex inferences about people's internal properties based on group membership; education about within- and between-group diversity as it relates to clothing is crucial to ensure the prevention of overgeneralization, stereotyping and prejudice.

Hendy, Basmah

POSTER TITLE: Nanoparticle-based CNS Drug Delivery to Improve the Safety and Efficacy of IRX4204 in the

Treatment of Schizophrenia

AUTHORS: Basmah Hendy, Gracious DS Kasheke, Antoine Hakim, Marianna Foldvari, George S Robertson

ABSTRACT: Schizophrenia is a psychiatric disorder characterized by positive, negative, and cognitive symptoms. There are no drugs that alleviate the negative and cognitive symptoms of schizophrenia - which have been linked to impaired lipid synthesis. IRX4204 is an experimental drug that increases lipid synthesis by activating the retinoid X receptor (RXR). However, IRX4204 suffers from poor brain penetration and the induction of hyperlipidemia by activating RXRs in adipose tissues. We propose that these problems can be overcome using nanoparticles (NPs) to preferentially deliver IRX4204 to the central nervous system (CNS). Unlike intravenous and oral drug administration, intranasally delivered drugs bypass the blood brain barrier and gain direct access to the CNS. Indeed, intranasal administration of IRX4204-NPs enhanced CNS drug uptake while reducing systemic drug exposure in mice. This increased the expression of RXR-induced genes in the CNS, indicating the enhanced CNS uptake of IRX4204 and release of this drug in sufficient amounts to activate RXRs. We now plan to determine whether our IRX4204-NPF is safe and effective in animal models of schizophrenia.

McCord, Betsy

POSTER TITLE: Structural Brain Abnormalities in Long-COVID: Impact on Cognition

AUTHORS: Betsabe McCord & Carlos R. Hernandez-Castillo

ABSTRACT: Introduction: Emerging evidence suggests that COVID-19 infection may induce long-term cognitive impairments. A detailed assessment of such impairments and their neurological underpinnings is still missing. We aim to identify associations between brain abnormalities and cognitive performance in long-COVID patients. Methods: We acquired MRI images from 42 long-COVID patients, and 37 healthy controls. Cognitive performance was assessed using two standardized test batteries (CBS and CCAS). Group differences on grey matter density (GMD) were analyzed using voxel-based morphometry (VBM). Mean GMD was extracted from clusters showing significant difference between groups. Statistical associations between GMD and cognitive test scores were assessed using false discovery rate correction to control for multiple comparisons. Results: VBM revealed significant differences in multiple brain regions, including the left precentral gyrus, left posterior cerebellum (Crus I), left pons, left anterior cerebellum, brainstem, and right anterior cerebellum (V & VI). Significant correlations were found between GMD and cognitive test scores, including semantic fluency, verbal recall, deductive reasoning, spatial planning, verbal short term and working memory, visuospatial working memory, and episodic memory. Conclusion: Our findings shows that GMD is associated with specific cognitive deficits in long-COVID patients. Highlighting the importance of understanding the neurological sequalae of the COVID-19 infection.

Matthews, Bianca

POSTER TITLE: Anti-Black Racism and Children's Pain Management: A Proposal

AUTHORS: Bianca Matthews, Dr. Samantha Louie-Poon, Dr. Christine Chambers

ABSTRACT: Introduction: Although all children will experience pain (acute or chronic), Black populations are disproportionately negatively impacted due to anti-Black racism. The objective is to gain an understanding of Black caregivers and their children's experiences of the impacts of anti-Black racism on their children's pain management. Methods: A narrative inquiry study, guided by critical race theory, will be used and it aims to capture the stories of lived experiences for marginalized populations. Five dyads (pairs of Black caregivers and their children between 12-18 years old) will be recruited. They must live in Canada and have had pain-related interactions with the Canadian healthcare system. Participants will take part in semi-structured interviews to capture their pain experiences. Interviews will be transcribed verbatim and inductively analyzed through thematic narrative analysis. Significance: Understanding diverse pain experiences of Black families will highlight research gaps in incorporating

patient perspectives on anti-Black racism in children's pain treatments.

Sozuer, Can

POSTER TITLE: Modeling the impacts of sleep history dependent structural plasticity of astrocytic processes on glutamatergic transmission at tripartite synapses to lateral hypothalamic orexin neurons

AUTHORS: Can Sozuer, Vikas Pandey, Yoshiyuki Kubota, Kazue Semba

ABSTRACT: Sleep is homeostatically regulated through interaction of sleep- and wake-promoting neurons such as orexin neurons, and the role of astrocytes in their synaptic interaction is an active area of investigation.Our previous work showed that sleep deprivation induces presynaptic inhibition of excitatory inputs to orexin neurons and that astrocytic processes withdraw from synapses after sleep deprivation. To determine whether astrocyte withdrawal can account for presynaptic inhibition, we computationally modeled synaptic release, the influence of glutamate clearance by astrocytes on glutamate diffusion in the extracellular space, and postsynaptic currents. Different conditions of astrocyte position, glutamate transporter density, and synapse geometry were compared to assess their effectiveness for inducing presynaptic inhibition through presynaptic mGluRs. The model elucidated that the astrocytic regulation on glutamate clearance is most effective under high-affinity presynaptic inhibition mechanisms such as that mediated by group III mGluRs found at synapses to orexin neurons. The results validate astrocyte withdrawal as a potential mechanism for our previous electrophysiological finding of sleep history dependent presynaptic regulation of wake-promoting orexin neurons.

Glenham, Cecilia

POSTER TITLE: T-Pattern Analysis of Grooming Behaviour in the Neurexin-1+/- mouse model of Autism Spectrum Disorder

AUTHORS: Cecilia Glenham

ABSTRACT: Autism Spectrum Disorder (ASD) is a neurodevelopmental disorder characterized by deficits in communication, social interaction, and repetitive behaviors. Mutations of the Neurexin (NRXN) gene result in disrupted synaptic function and have been strongly implicated in ASD, although the precise mechanisms remain unclear. Mouse self-grooming is a repetitive, sequentially patterned behavior model, making it an ideal translational tool for ASD research. The Neurexin1+/- (Nrxn1+/-) is a novel mouse model of ASD, and our study was the first to perform grooming research for this strain. We previously examined grooming microstructure in male and female Nrxn1+/- mice compared to wild-type (Nrxn1+/+) mice by using traditional quantitative analysis. This study is a continuation of this research and by the implementation of T-Pattern analysis to detect hidden patterns of behavior. Preliminary results show that genotypic and sexually demographic differences in grooming patterns may exist between Nrxn-1+/ mice. This research helps enhance our understanding of the microstructural organization of repetitive behaviors in ASD models and highlights the advantages of using T-patterns in behavioral analysis.

Aihara de Oliveira, Letícia

POSTER TITLE: ADHD and Pedestrian Behaviour

AUTHORS: Letícia Aihara de Oliveira, Alessandra Bianchi

ABSTRACT: Survival in traffic requires proper functioning of various psychological functions, related deficits, such as attention deficit hyperactivity disorder (ADHD), can be related to greater risk-taking behavior in traffic. The aim of this study was to investigate the relationship between ADHD and risk-taking behavior in pedestrians. Participants were asked to complete a questionnaire on pedestrian behavior, and the Attention Deficit Hyperactivity Disorder Scale for Adults. The sample consisted of 305 adults aged between 18 and 59, literate, and of both sexes. 36.3%

were classified as having significant inattention compatible with severe difficulties, and 29.9% were classified as having a moderate level. There was a statistically significant positive correlation between ADHD-related symptoms and pedestrian behavior risk, which means that the higher level of ADHD, the greater the risky behavior as a pedestrian.

McDonald, Emily

POSTER TITLE: More Absolute Moderate-to-Vigorous Physical Activity is Associated with Better Health Related Quality of Life in Outpatients with Acquired Brain Injuries

AUTHORS: Emily E. MacDonald[1†], Liam P. Pellerine[2†], Katerina Miller[3], Ryan J. Frayne[2], Myles W. O'Brien[4,5*]. Authors contributed equally.

1 Department of Neuroscience, Dalhousie University, Halifax, Nova Scotia, Canada. B3H 4R2

2 Division of Kinesiology, School of Health and Human Performance, Faculty of Health, Dalhousie University, Halifax, Nova Scotia, Canada. B3H 4R2

3 Acquired Brain Injury NeuroCommons, Nova Scotia Health Authority, Halifax, Nova Scotia, Canada

4 Department of Medicine, Université de Sherbrooke, Sherbrooke, Quebec, Canada

5 Centre de Formation Médicale du Nouveau-Brunswick, Université de Sherbrooke, Moncton, New Brunswick, Canada

ABSTRACT: Health-related quality of life (HRQoL) is a patient perceived measure of physical, social, and emotional health. An acquired brain injury (ABI) occurs due to damage to the brain after birth, and individuals typically present with lower HRQoL. Most activity research characterizes moderate-vigorous physical activity (MVPA) in absolute terms; however, relative physical activity levels have been promoted. We test if higher MVPA duration in absolute/relative levels is associated with higher HRQoL in outpatients with an ABI. Twenty-six adults (54±13 years, 16 female) with an ABI completed the Quality of Life After Brain Injury questionnaire, a six-minute walk test (measure of aerobic fitness; 490±105 m) and wore an activPAL 24 hr/d for 7 days. Participants had an average HRQoL score of 53.4±15.0 (out of 100). Absolute MVPA (74.6±91.0 min/week, b=0.09, p=0.03), but not total (565.7±264.8 min/week, p=0.47), was associated with HRQoL. Neither relative LPA (521.4±244.9) nor relative MVPA (33.5±34.9 min/week) were associated with HRQoL (both, p>0.14). Targeting more absolute MVPA, but not necessarily relative MVPA, may be an effective strategy for interventions aiming to improve HRQoL for those with ABIs.

Reeves, Emma

POSTER TITLE: Children's Beliefs About L2 Learners and Learning Motivations

AUTHORS: Emma Reeves, Rebeka Workye, Dr. Drew Weatherhead

ABSTRACT: Although past research has examined children's attitudes and beliefs about second language (L2) learners, very little has investigated the role of motivation. We examined whether an L2 learner's motivations for learning a language impacts children's social inferences. Participants aged 5-8 (N=90) were introduced to two fictional L2 learners possessing different learning motivations (social or academic) then given a selective learning task and asked which character they believed had higher L2 proficiency. On average, participants preferred to ask the social character for the name of an object but equally endorsed object names by both characters. Additionally, when asked about proficiency, participants chose characters equally. Overall, these results demonstrate that children have varied preferences for L2 learners depending on their learning motivation.

Schwenck, Grace

POSTER TITLE: Fear of failure intensifies the relationship between statistics test performance and self-esteem: A two-group pre-post experiment

AUTHORS: Gracielle C. Schwenck, BSc[1], Sean M. Alexander, BSc[2], Robert A. Cribbie, PhD[3], Gordon L. Flett, PhD[3], Aaron Shephard, MSc[1] & Sean P. Mackinnon, PhD[1]

1Dalhousie University
2 Carleton University
3 York University

ABSTRACT: Statistics courses are common within post-secondary programs, and many students report significant cognitive and affective barriers to statistics education. In line with the vulnerability-stress model, students with higher fear of failure may be more likely to experience decreases in state self-esteem (e.g., performance and social self-esteem) following the stressor of academic failure. This study examined changes in students' state self-esteem following a failed statistics test and identified if this association was moderated by fear of failure. Post-secondary students (N = 329) participated in a two-group, pre-post, between-subjects experimental study. Participants were randomly assigned to receive an easy or hard 5-item statistics test and feedback on their final score. Analyses used moderated multiple regressions. Students in the hard test condition had greater decreases in post-test performance and social self-esteem. Fear of failure moderated this association for performance self-esteem, but not social self-esteem. Results suggest students high in fear of failure are more likely to experience aversive responses to failure in a statistics assessment. Thus, interventions aimed at adjusting cognitions surrounding failure (e.g., reframing failures as learning opportunities) may prevent procrastination or unenrolment from challenging courses.

Boossom, Isabella

POSTER TITLE: The Impulsivity Subscale of The Substance Use Risk Profile Scale (SURPS) Has Predictive Utility for Binge Drinking and Binge Eating Among Undergraduates Over Time

AUTHORS: Simon B. Sherry; Aislin R. Mushquash; Christopher J. Mushquash; Daniel S. McGrath; Sherry H. Stewart

ABSTRACT: The Substance Use Risk Profile Scale (SURPS) is a measure that is used to select youth for personality-targeted interventions (i.e., PreVenture) that have been found to prevent or reduce risky substance use in young people. We tested whether the SURPS impulsivity subscale would predict changes in binge drinking and binge eating over time. Undergraduates (N = 410, M age = 20.7 years, SD = 5.5, women n = 307, men, n = 103) from two universities completed a longitudinal study with two weeks separating baseline and follow-up. At baseline, participants completed the SURPS impulsivity subscale and binge drinking and binge eating measures. At follow up, participants completed the measures again. Linear regression analyses showed that SURPS impulsivity (B = .12, p = .02) significantly predicted binge drinking and binge eating at follow up. Findings reveal that the SURPS impulsivity subscale is a useful tool in predicting increases in binge behaviour.

Whyte, Kameryn

POSTER TITLE: Uses of broad consent among diverse populations: A scoping review protocol

AUTHORS: Kameryn Whyte, Debbie Johnson Emberly, Shannon Johnson

ABSTRACT: The purpose of this scoping review is to examine the use of broad consent among vulnerable populations in Canada, with the intention of identifying cultural considerations for incorporation in the future implementation of broad consent in research. There are two research questions in this review: (1) Have there been any uses of broad consent among diverse populations in Canada? (2) What are the cultural considerations that need to be incorporated in the implementation of broad consent for future research? The diverse populations under study are: Black/African Nova Scotian, Indigenous, 2SLGBTQI+, and Newcomers. They have been identified as the priority population at the Centre for Psychological Health, a PhD clinical psychology training clinic. The findings will contribute to the identification of future areas of research and guidance for the development of policy related to obtaining broad consent for future research. We will share the development of the protocol, including

inclusion/exclusion criteria, search strategy, data extraction and analysis.

Collings, KC

POSTER TITLE: Seasonal change in neurogenesis within the vocal control system of wild black-capped chickadees (Poecile atricapillus)

AUTHORS: KC Collings, Broderick M. B. Parks, Leslie S. Phillmore

ABSTRACT: Birds are remarkable models in neuroscience as they exhibit neuroplasticity throughout their brain and throughout their lifetime. Temperate-zone songbirds often show seasonal changes in the neural structures supporting vocal behaviour, including changes in neurogenesis. These seasonal differences in the brain are believed to reflect seasonal differences in vocal behaviour. Here, we evaluate neurogenesis in the vocal control system of wild male and female black-capped chickadees (Poecile atricapillus) across spring, summer and winter. Neurogenesis was quantified by measuring the immunolabelling of doublecortin (DCX), in vocal control nuclei HVC, Area X, and adjacent (non-nuclear) controls. We found that Area X had significantly more neurogenesis than its two control regions and HVC, while HVC did not differ from its controls. Birds captured in winter had significantly more neurogenesis in HVC than birds captured in spring, and we did not find seasonal differences in Area X. We did not find any sex differences in neurogenesis which may reflect the relatively small sex difference in vocal behaviour of chickadees. Overall, our data suggest the recruitment of new neurons to two distinct vocal control nuclei reflects the seasonally changing (or unchanging) demands of distinct components of vocal behaviour in this species.

Roddick, Kyle

POSTER TITLE: Discovery of a new mouse model for spontaneous late-onset epileptic seizures

AUTHORS: Kyle Roddick, Richard E. Brown, Wyatt Ortibus, Hong Lu and Ann Marie Craig

ABSTRACT: More people over 75 years of age develop epilepsy than children under 10. (Sen et al., 2020. Lancet 395:735-748). We have discovered a novel mouse model that (1) spontaneously develops seizures with no external induction; and (2) develops these seizures at an older age. We tested mouse lines with mutations in the neurexin-1 synaptic organizing protein: Nrxn1+/- mice with one allele disrupted, Nrxn1 ΔS5/ΔS5 mice with altered splice site 5 (Hong et al., 2023. Cell Reports 42:112714), Nrxn1 ΔS5/- mice, and WT control mice. Mice show seizures starting at 11 months of age. Seizures were scored on a revised Racine scale. Young mice showed elevated excitatory but not inhibitory synaptic activity in the Nrxn1 ΔS5/ΔS5 mice (Hong et al., 2023). We continue to screen the mice in our colony for seizures and plan to examine the neuroanatomical and neurochemical basis of these seizures.

Rahman, Lamisa

POSTER TITLE: Mechanisms Localizing NMDA Receptors to Synapses

AUTHORS: Rahman, L.; Ferguson, L.; Krueger, S. R.

ABSTRACT: NMDA-type glutamate receptors (NMDARs) play a key role in synaptic plasticity, and their properties are determined by their subunit composition. Studies addressing the regulatory mechanism of NMDAR recruitment to synapses have focused on protein interactions with the cytoplasmic C-terminal domains (CTD) of GluN2 subunits, whereas potential interactions of extracellular domains are unexplored. In this study, we examined the relative roles of the extracellular N-terminal domain (NTD) and the CTD of GluN2A subunits in the synaptic positioning of NMDARs by expressing terminal deletion mutants in rat hippocampal neurons. The constructs were tagged with super-ecliptic pHluorin to detect plasma membrane-localized NMDARs. While NMDARs containing full-length GluN2A were found to be significantly enriched at axospinous synapses, neither deletion mutants lacking the CTD nor the NTD exhibited preferential synaptic localization. Our results support our hypothesis that protein interactions of the

CTD and the NTD are necessary for the synaptic localization of GluN2A-containing NMDA receptors.

Dang, Lan

POSTER TITLE: Mice change their mind in a two-alternative-forced-choice psychophysical task.

AUTHORS: Dang, L.T.J., Michaud, N.M., Crowder, N.A.

ABSTRACT: Decision making involves gathering and evaluating information before committing to an action. However, even after making the decision, we continue processing information, potentially leading to reconsiderations, or changes of mind (CoM). In this study, we investigated the decision-making processes of mice using a two-alternative-forced-choice contrast detection task, where mice used a steering wheel to indicate the side of a grating stimulus. By continuously monitoring the steering wheel position as a choice was being registered, we found that CoM occurred more frequently in difficult trials. Moreover, mice tended to change to the correct choice on easy trials, but CoM direction approached chance on the most difficult trials. These results highlight several parallels in decision-making processes between humans and a genetically tractable animal model.

Elliott, Laura

POSTER TITLE: Investigating fNIRS Test-Retest Reliability during Lexical Decision

AUTHORS: Laura Elliott, Saisha Rankaduwa, Devon Bode, Marilla Hulls, Clare Reynolds, Cindy Hamon-Hill, Aaron Newman

ABSTRACT: Functional near infrared spectroscopy (fNIRS) is a new technique in the field of neuroimaging, and the reliability of its signals has not been established during reading tasks. Prior to using fNIRS to investigate reading, it is necessary to establish the technique provides a reliable measure of task-related neural activity. The objective of the present study is to assess the test-retest reliability (TTR) of the signals recorded by fNIRS, during a lexical decision task, at the group and subject level. We recruited adults to complete a lexical decision task during fNIRS recording twice, with sessions occurring one week apart. Contrary to our hypothesis, preliminary results have demonstrated no significant print, sublexical, or lexical tuning effects at the group level. Additionally, our results demonstrate significant variability in TTR at the single subject level, for all tuning effects. At the group level, we observed fair to good TTR for print and lexical tuning, but not for sublexical tuning. Finally, preliminary group level results have demonstrated better TTR when analyzing data at the level of regions of interest, relative to individual channels. Identifying the reliability of fNIRS during a standard reading task will aid inguiding recommendations in its use in future studies of reading.

Aubin, Lindsay

POSTER TITLE: Predictors of Sexual Arousal From Exhibitionistic Behaviour

AUTHORS: Lindsay Aubin & Dr. Julie Blais

ABSTRACT: This study investigated self reported measures of hypersexuality, grandiose narcissism, and sexual entitlement as predictors of arousal from exhibitionism. An anonymous self-report Qualtrics survey was completed by 1,200 participants who were at least 18 years of age, living in Canada, and able to read English. All three predictors were found to be moderately positively correlated with exhibitionism, with sexual entitlement having the strongest positive correlation ($r = .32$), followed by hypersexuality ($r = .30$), and grandiose narcissism ($r = .23$). A multiple linear regression with all 3 predictors entered was significant, predicting 13.90% of the variance in sexual arousal from exhibitionism. Gender differences were found, with men scoring significantly higher than women on all measures with effect sizes ranging from 0.20 (exhibitionism) to 0.45 (hypersexuality). This preliminary study makes important contributions to an area of sexual offending that has very little research and sheds light on gender

differences in the prediction of arousal from exhibitionism. Keywords: exhibitionism, hypersexuality, sexual entitlement, grandiose narcissism, predictor, motivation

ten Haaf, Mark

POSTER TITLE: The (In)consistency of Unilateral Optogenetic Modulation for Contrast Detection in Transgenic Mice

AUTHORS: ten Haaf, M.R., Michaud, N.M., Crowder, N.A

ABSTRACT: Mice are a valuable model in neuroscience research as they can be used with a variety of genetic tools to link brain and behavior. One such method is optogenetics, where specific neural subtypes express light sensitive channels that can be modulated with high temporal precision. This study used optogenetics to hyperpolarize somatostatin-expressing-interneurons unilaterally in primary visual cortex (V1) while mice completed a two-alternative-forced-choice contrast detection task. We found that for a subset of animals the unilateral disinhibition of V1 caused a side bias in the behavioral task. Here we explore the stability of this side bias over multiple weeks of testing.

Raja, Moomina

POSTER TITLE: Sleep for health in hospital and at home: The need to prioritize children with autism spectrum disorder and other neurodevelopmental disabilities

AUTHORS: Momina Raja, Mitchell Mathieson, Victoria Foxall, Isabel M. Smith, and Megan Thomas

ABSTRACT: Healthy sleep is essential for development and protects against chronic pain and mood issues, important factors particularly for children with neurodevelopmental disabilities (NDDs) like autism. Over 80% of these children experience sleep difficulties, which are exacerbated during hospitalization, affecting both children and their parents. Our study assessed sleep disruptions and potential solutions for hospitalized children and their parents. Sleep, activity, mood, and environmental data was collected from eight parent-child (3 to 11 years) dyads over 48 hours in hospital and at home. Five parents also completed interviews post-discharge. Thematic analysis identified five themes: importance of sleep, influences on sleep, managing challenges, support systems, and the potential for positive change. Excessive noise and minimal light diurnal variation were recorded. Children with NDDs had more frequent and longer hospitalizations. Families proposed solutions, and increased awareness improved sleep quality. A follow-up study in partnership with families with NDDs and clinicians is currently underway.

Dockrill, Mya

POSTER TITLE: Evaluating the usability and feasibility of the ABCs of SLEEPING for Adults mobile application: A proposal

AUTHORS: Dockrill, M., Bertrand, M., & Corkum, P.

ABSTRACT: Good sleep health plays a vital role in adults' overall health and well-being; however, access to evidence-based sleep health resources is limited. The current study aims to assess the usability and feasibility of the ABCs of SLEEPING for Adults mobile app which provides evidence-based recommendations to improve sleep health based on an assessment of current sleep habits. Twenty adults will a) review and use the app, b) participate in an interview where they will provide feedback on the usability of the app by answering questions based on Morville & Sullenger's (2010) "User Experience Honeycomb" framework, and c) provide feedback pre- and post-intervention via questionnaires to assess the app's feasibility (including its implementation and preliminary effectiveness). The feedback from participants who reviewed and used the app will be incorporated into a modified version. This app has the potential to be an accessible resource for adults to promote sleep health, and if deemed usable and feasible, our next steps will be to test the full-scale effectiveness.

Singh, Prem

POSTER TITLE: Sst-Cre; Ai40D transgenic mice performing a psychophysical size discrimination task do not show sex- or cohort-related differences in task accuracy.

AUTHORS: Singh, P.K., Michaud, N.M., ten Haaf, M.R., Nadeau, E., Atwell, I., Dang, L.T., Adesalu, K., Crowder, N.A.

ABSTRACT: Mice serve as an important animal model in neuroscience because of their compatibility with various genetic techniques. Our research program investigates links between neural circuits in primary visual cortex (V1) and perception in transgenic mice using optogenetic tools, which modify the activity of specific neuron subtypes using light. We are evaluating size perception in mice using a two-alternative-forced-choice size discrimination task where mice turn a steering wheel to bring the larger of the two stimuli towards to centerline for a reward. During certain trails, we illuminated V1 bilaterally with yellow light to suppress Archaerhodopsin expressing somatostatin-positive interneurons. Our present analysis of the psychophysical data focused on possible sex and cohort differences, accounting for other factors beyond optogenetic manipulation that could affect task performance.

Garay, Pristine

POSTER TITLE: Exploring Behavioral Consequences in Cux2iresCre mice: Implications in Neurodevelopmental Disorder Research

AUTHORS: Pristine Garay, Dr. Emily Witt, Dr. Danielle Stanton-Turcotte and Dr. Angelo Lulianella

ABSTRACT: Neurodevelopmental disorders (NDDs) affect social and repetitive behaviours differently in males and females, highlighting the need to understand the underlying genetic mechanisms. The gene Mllt11, previously studied in cancer, is now linked to neurodevelopment as conditional knockout (cKO) of Mllt11 in the cerebral cortex of mice reflects NDD-like traits. However, this model involves the Cre-loxP system, which requires the insertion of the Cre transgene upon gene inactivation that may cause additional behavioural changes. As such, we conducted a behavioural analysis of Cux2iresCre mice to assess the Cre transgene's impact on NDD characteristics compared to wild-type mice. No significant differences were found in weight regulation, olfactory function, and most anxiety-like and repetitive behaviours. However, sex-dependent differences emerged: male Cux2iresCre mice showed slight social scent response impairment, while females exhibited decreased chamber entries, increased social preferences, and digging behaviour. These findings underscore the sex-linked neurobehavioral effects of the Cre transgene in NDD research.

Lewis, Ry

POSTER TITLE: Exploring the development of the N400 event-related potential in new learners of Spanish words

AUTHORS: Ry Lewis, Aaron Newman, Mobina Ali

ABSTRACT: The N400 ERP increases in amplitude in response to semantic violations. Learners of an unfamiliar language (UL) show no N400 ERP in response to semantic violations with UL words before training. Previous work showed that after 10 days of training, learners develop an N400 effect for semantic violations using the UL words. The current study was interested in exploring the development of an N400 for learners of an UL by performing EEG recordings throughout the entire training process, rather than simply at the end. Participants learned the meaning of Spanish words through association with images over a single session. Participants showed significant gains in correctly reporting whether an image and word matched or mismatched in meaning from before to after training, with proportion correct approaching ceiling. However, no significant N400 effects were found between match and mismatch test conditions from before to after training when averaged across participants. More days of training are likely necessary for the N400 to reliably emerge in participants.

Hawkins, Sam

POSTER TITLE: Perfectionism and Post-Secondary-Specific Stressors as Risk Factors for Academic. Outcomes: A Vulnerability-Stress Model

AUTHORS: Sam Hawkins, Taylor Hill & Sean Mackinnon

ABSTRACT: Introduction: Self-critical perfectionism and stress have been implicated as risk factors for maladaptive academic outcomes (i.e., decreased academic self-efficacy, decreased academic self-concept, and increased academic burnout). In the present study, we investigated a vulnerability-stress model, testing whether perfectionism (self-critical and rigid) moderates the relationship between stressor severity (academic and interpersonal) and academic outcomes.
Method: A sample of 384 post-secondary students completed a cross-sectional survey involving questionnaires assessing perfectionism, stress, academic self-efficacy, academic burnout, and academic self-concept. Results: Main effects of stressor severity (both academic and interpersonal) were found to predict decreased academic self-efficacy and academic self-concept, as well as increased academic burnout. Main effects of self-critical perfectionism were found to predict decreased academic self-efficacy and increased academic burnout. Main effects of rigid perfectionism generally predicted increased academic self-efficacy and increased academic self-concept when controlling for stressor severity. None of the proposed interaction effects were statistically significant.
Discussion: Findings suggest stressor severity (both academic and interpersonal) and self-critical perfectionism are strong predictors of maladaptive academic outcomes, and risk factors for poor academic functioning. The lack of an interaction effect suggests that a vulnerability-stress model does not explain why perfectionism leads to maladaptive academic outcomes.

Ritsema, Vanessa

POSTER TITLE: Accent Intelligibility: Understanding the effects of race-based expectations and linguistic background

AUTHORS: Ritsema, V., Woryke, R., & Weatherhead, D.E.

ABSTRACT: Race-based expectations and linguistic background can both influence accent intelligibility. However, past research on accent intelligibility has yielded conflicting results, and the underlying mechanisms of accented-speech processing remain unclear. Furthermore, the effects of race-based expectations and linguistic background have been studied largely in isolation, with past research focusing on monolingual listeners. The current study seeks to better understand the combined effects of race-based expectations and linguistic background on accent intelligibility by replicating the experimental design of McGowen (2015) with bilingual participants. The current study recruited 127 American participants fluent in English and Norwegian who listened to and transcribed 60 Mandarin-accented English sentences in noise. Sentences were presented with a White face or an East Asian face, with each condition designed to match or violate listeners' race-based expectations. Following sentence transcription, participants were asked to rate how strong they found accent to be from the experiment on a Likert scale. Participants also completed a short questionnaire that assessed their language usage, proficiency, familiarity, and attitudes. For analysis, a generalized linear mixed effects model was fitted to the data with condition, sex, accentedness rating, and participant language use as predictor variables. Results suggested that speaker race is not a salient speaker cue for bilingual listeners, and that bilingual listeners may therefore have different associations between ethnicity and accent compared to monolingual listeners.